Smart Soil Detection & Crop Recommendation Report

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Soil Parameters

pH	Nitrogen (N)	Phosphorus (P)	Potassium (K)
8.1	245	18	310

Weather Conditions

Temperature (°C)	Humidity (%)	Rainfall (mm)
26.5	72	110

Soil Health Score

32 / 100 — Poor

pH	N	P	K
17.5/25	0/25	15/25	0/25

Recommended Crops

#	Crop	Confidence
1	Grapes	17.5%
2	Coffee	17%
3	Watermelon	16%

Fertilizer Advice

Nitrogen (High): Reduce nitrogen input; avoid urea top-dressing — N=245.0 exceeds the ideal band (50.0-100.0); risk of lodging/leaching.

Phosphorus (Low): DAP (18-46-0) or Single Super Phosphate — P=18.0 is below the ideal band (30.0-60.0).

Potassium (High): Avoid further potash application this season — K=310.0 exceeds the ideal band (30.0-60.0).

Soil Improvement Advice

- Soil is alkaline — apply gypsum or elemental sulfur, and add organic matter.
- Incorporate bone meal or rock phosphate to improve long-term phosphorus availability.
- Practice crop rotation and cover cropping to sustain long-term soil structure and microbial health.

Risk Alerts

No significant risks detected for the given conditions.

This report is generated by an automated decision-support system based on the soil and weather values provided. Image-based soil classification is shown for reference only and is not used for crop or fertilizer decisions. Please consult a local agronomist before making major input investment decisions.